Artefact 2: Group Assignment – Airbnb Pricing Project

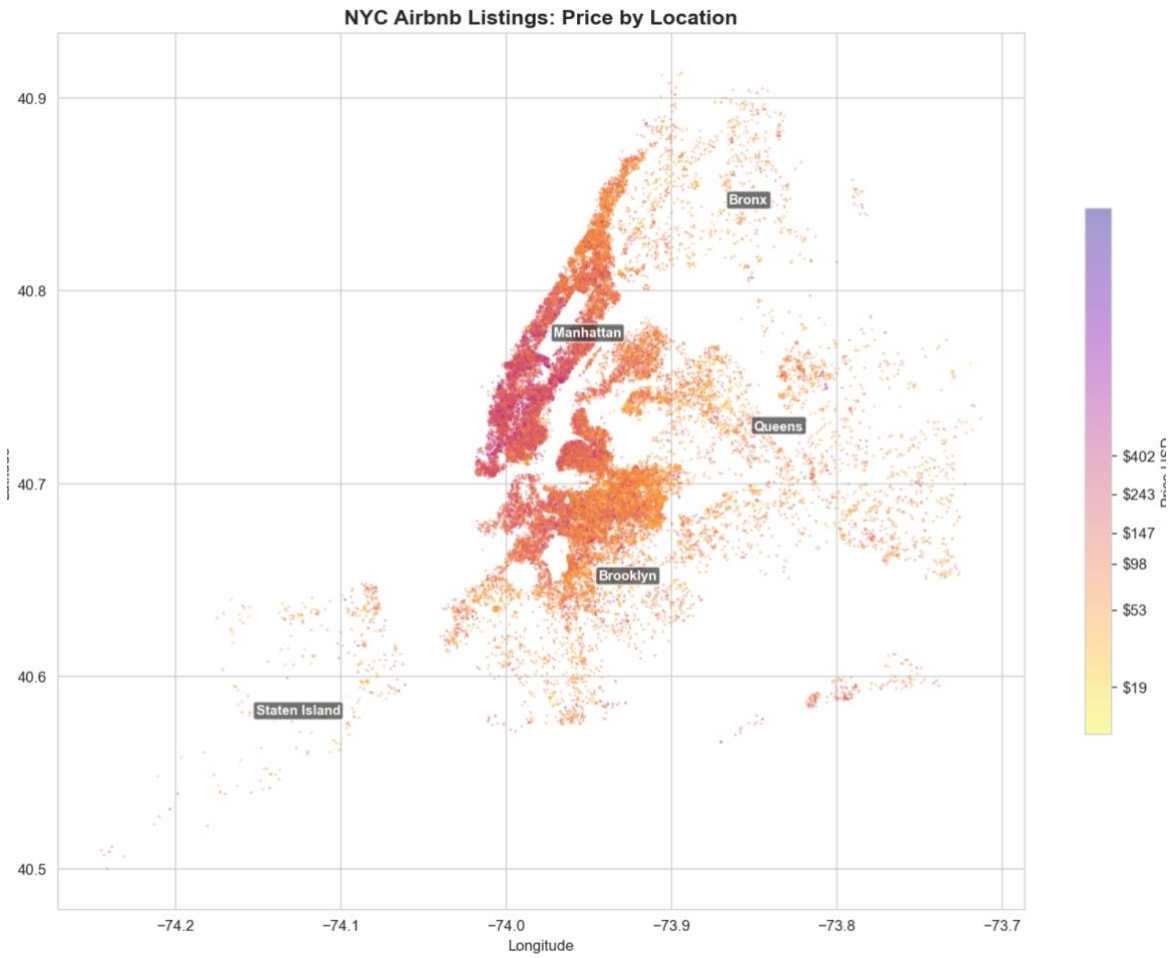
Final Group Report Link: [Report - Final.pdf]
Dataset overview / Descriptive Statistics: First 5 columns

	id	host_id	latitude	longitude	price
count	48895.00	48895.00	48895.00	48895.00	48895.00
mean	19017143.24	67620010.65	40.73	-73.95	152.72
std	10983108.39	78610967.03	0.05	0.05	240.15
min	2539.00	2438.00	40.50	-74.24	0.00
25%	9471945.00	7822033.00	40.69	-73.98	69.00
50%	19677284.00	30793816.00	40.72	-73.96	106.00
75%	29152178.50	107434423.00	40.76	-73.94	175.00
max	36487245.00	274321313.00	40.91	-73.71	10000.00

Price distribution plot: Heavily right-skewed distribution

Price by neighbourhood group: boxplot without outliers

Geographical Price Distribution: Log1p transformation is applied to price before colouring and compressing the scale so that luxury outliers do not dominate and variation across the majority of listings remains visible.



This artefact represents the application of machine learning to a real-world business problem, focusing on Airbnb pricing in New York City. It includes the final group report and selected outputs from the exploratory data analysis and preprocessing stages, which were my primary areas of contribution.

This work demonstrates my ability to prepare and critically assess datasets prior to modelling. I was responsible for analysing dataset structure, handling missing values, and identifying inconsistencies such as zero-price entries. These preprocessing decisions were essential in ensuring data quality and directly influenced the reliability of downstream modelling.

Through exploratory data analysis, I identified key patterns that shaped the overall project direction. In particular, I observed that price distribution was highly skewed, requiring transformation, and that categorical variables such as location and room type had a significantly stronger impact on price than numerical features. Correlation analysis further showed that most numerical variables had weak relationships with price, highlighting limitations in predictive power.

Although I was not directly responsible for implementing the regression and clustering models, engaging with the full project allowed me to critically interpret the results. I recognised that the regression model captured general trends but struggled with local variation, and that clustering provided useful segmentation but did not fully represent the continuous nature of the market.

This artefact also reflects my experience working within a development team. I contributed to the analytical foundation of the project and supported the overall interpretation of results, while collaborating with team members responsible for modelling. This strengthened my ability to work in a structured, role-based environment and understand how different stages of a machine learning pipeline connect.